

Role of Member	Member	Test Case ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
Booking Management	M1	TC-BM-01	GA Route Generation	Verify GA generates a valid sequence for 5 locations	Locations exist in DB.	1. Select 5 spots. 2. Run GA.	List of 5 Spot IDs.	Non-repeating sequence of all spots.	Generated sequence [1,4,2,5,3].	PASS
	M1	TC-BM-02	Time Window Logic	Ensure route respects opening hours.	Spot hours set in DB.	1. Set start time 08:00. 2. Check arrival.	Start: 08:00 AM.	Arrivals occur within Open/Close time.	All arrivals within valid hours.	PASS
	M1	TC-BM-03	Fitness Improvement	Verify GA fitness improves over generations.	GA parameters set.	1. Run for 50 gens. 2. Compare scores.	Generation Limit: 50.	Final Fitness < Initial Fitness.	Fitness reduced from 450 to 120.	PASS
	M1	TC-BM-04	Booking Update	Re-run optimization on date change.	Existing booking.	1. Change date. 2. Re-optimize.	New Date: 2026-05-20.	New optimized itinerary generated.	Itinerary successfully updated.	PASS
	M1	TC-BM-05	Late Start Edge Case	Handle start time after closing hours.	Locations close at 18:00.	1. Set start 19:00. 2. Run GA.	Start: 19:00.	Error: "No valid route".	Alert: "Cannot generate route".	PASS
Payment Management	M2	TC-PM-01	Redirect Speed	Verify redirection is under 2 seconds.	Payment Gateway active.	1. Click Pay. 2. Measure time.	Payment Amt: 5000 LKR.	Redirection occurs < 2.0s.	Redirected in 1.4s.	PASS
	M2	TC-PM-02	Status Sync	Update status from Pending to Success.	DB connection active.	1. Complete payment. 2. Check DB.	Transaction ID: TXN001.	DB Status updated to 'SUCCESS'.	DB updated to 'SUCCESS'.	PASS
	M2	TC-PM-03	Currency Conversion	Test non-LKR currency input.	Exchange API active.	1. Input USD. 2. Check conversion.	Input: \$50 USD.	Correct LKR equivalent calculated.	Converted to LKR 15,000.	PASS
	M2	TC-PM-04	Decline Handling	Simulate declined card transaction.	Sandbox mode active.	1. Use 'Declined' card. 2. Check UI.	Card: 4000...0002.	UI shows Error: "Declined".	"Transaction Declined" shown.	PASS
	M2	TC-PM-05	Data Masking	Ensure card details aren't logged.	Logging enabled.	1. Enter card. 2. Check server logs.	Card: 4111...1111.	Logs show masked values (****).	Logs show "XXXX-XXXX".	PASS
Tour Package Selection	M3	TC-GS-01	Guide Matching	Match user language to guide skills.	Guides in DB.	1. Set Pref: Sinhala. 2. Filter guides.	Language: Sinhala.	List of Sinhala speaking guides.	3 guides found/matched.	PASS
	M3	TC-GS-02	Schedule Conflict	Check for overlapping activities.	Guide schedule loaded.	1. Book overlapping slot. 2. Check validation.	Slot: 10:00 - 12:00.	Error: "Schedule Conflict".	Blocked: "Guide is busy".	PASS
	M3	TC-GS-03	Package Pricing	Verify total updates with add-ons.	Price rules set.	1. Select add-on. 2. Check total.	Add-on: Luxury Transport.	Total = Base + Add-on price.	Total calculated correctly.	PASS
	M3	TC-GS-04	Guide Availability	Check guide status for specific date.	Booking records exist.	1. Pick date. 2. Check status.	Date: 2026-04-10.	Status displays as 'Available'.	Displayed as 'Available'.	PASS
	M3	TC-GS-05	UI Consistency	Check Guide Profile layout standards.	CSS loaded.	1. View profile. 2. Check labels.	Profile ID: G001.	Standard layout and fonts used.	Layout aligns with UI specs.	PASS
Tourist Spot Browsing	M4	TC-SB-01	Semantic Accuracy	Search "Quiet nature" finds gardens.	Sentence Transformer active.	1. Input query. 2. Check top result.	Query: "Quiet nature".	Top result is Botanical Garden.	Top result: Peradeniya Garden.	PASS
	M4	TC-SB-02	Vector Storage	Verify embedding conversion.	Vector DB connected.	1. Add new spot. 2. Check Vector ID.	Spot: "Ancient Ruins".	Valid vector generated.	384-dim vector stored.	PASS
	M4	TC-SB-03	Empty Search	Handle nonsense keywords gracefully.	Search service running.	1. Input "xyz123". 2. Search.	Query: "xyz123".	Message: "No spots found".	"No results found" displayed.	PASS
	M4	TC-SB-04	Image Clarity	Verify high-quality image rendering.	Media server online.	1. Load spot page. 2. View photo.	Spot ID: S702.	Clear, non-lagged HD photo.	Image rendered in HD.	PASS
	M4	TC-SB-05	Tag Filtering	Filter by 'Historical' category.	Spots tagged in DB.	1. Select 'Historical'. 2. Check list.	Filter: Historical.	Only historical spots displayed.	List filtered correctly.	PASS
Feedback System	M5	TC-FR-01	Sentiment Score	Classify "Amazing trip" as Positive.	NLP model loaded.	1. Input review. 2. Run analysis.	"Amazing trip!".	Sentiment score > 0.8.	Sentiment Score: 0.92.	PASS
	M5	TC-FR-02	Summarization	Summarize 10 reviews into bullets.	10 reviews in DB.	1. Trigger summary. 2. Check length.	10 review objects.	Max 3 concise bullet points.	3 bullets generated correctly.	PASS
	M5	TC-FR-03	Profanity Block	Input restricted words in review.	Filter list active.	1. Enter bad word. 2. Submit.	Input: "[Profanity]".	Review Blocked or Flagged.	"Contains restricted words".	PASS
	M5	TC-FR-04	Average Rating	Verify star rating recalculation.	Existing ratings.	1. Add 5-star review. 2. Check Avg.	Rating: 5.0.	Global average rating increases.	Avg updated: 4.2 to 4.3.	PASS
	M5	TC-FR-05	Empty Validation	Submit review with no text.	Validation active.	1. Leave blank. 2. Submit.	Input: "".	Error: "Field cannot be empty".	Alert: "Please write a review".	PASS
User Authentication	M6	TC-UA-01	Chatbot Intent	Ask for travel advice.	Chatbot active.	1. Type "Help me plan". 2. Check reply.	"Help me plan".	Ask for interests or budget.	"Sure! What is your budget?".	PASS
	M6	TC-UA-02	Preference Tracking	Save interest from chat to profile.	Profile DB active.	1. Tell bot "I like hiking". 2. Check profile.	String: "I like hiking".	Profile database adds 'Hiking'.	Hiking added to preferences.	PASS
	M6	TC-UA-03	JWT Security	Verify token on successful login.	Auth service active.	1. Login valid user. 2. Check header.	User: test@example.com	Valid Bearer token returned.	JWT Token received.	PASS
	M6	TC-UA-04	Short Password	Block passwords < 8 chars.	Regex validation on.	1. Enter "123". 2. Register.	Pwd: pass	Error: "Password too short".	Error displayed correctly.	PASS
	M6	TC-UA-05	Session Timeout	Auto-logout after inactivity.	Timeout = 30 mins.	1. Wait 31 mins. 2. Refresh.	Inactivity: 31m.	Redirect to Login page.	Redirected to /login.	PASS